

Packet Tracer - Configure Router-on-a-Stick Inter-VLAN Routing

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.30	172.17.30.1	255.255.255.0	N/A
PC1	NIC	172.17.10.10	255.255.255.0	172.17.10.1
PC3	NIC	172.17.30.10	255.255.255.0	172.17.30.1

Objectives

Part 1: Add VLANs to a Switch

Part 2: Configure Subinterfaces

Part 3: Test Connectivity with Inter-VLAN Routing

Scenario

In this activity, you will configure VLANs and inter-VLAN routing. You will then enable trunk interfaces and verify connectivity between VLANs.

Instructions

Part 1: Add VLANs to a Switch

Step 1: Create VLANs on S1.

Create VLAN 10 and VLAN 30 on **S1**.

Step 2: Assign VLANs to ports.

a. Configure interfaces F0/6 and F0/11 as access ports and assign VLANs.

- Assign the port connected to **PC1** to VLAN 10.
- Assign the port connected to **PC3** to VLAN 30.

b. Issue the **show vlan brief** command to verify VLAN configuration.

S1# **show vlan brief**

```

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18

```

			Fa0/19, Fa0/20, Fa0/21, Fa0/22
			Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	Fa0/11
30	VLAN0030	active	Fa0/6
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Step 3: Test connectivity between PC1 and PC3.

From **PC1**, ping **PC3**.

Were the pings successful? Why did you get this result?

The VLANs cannot communicate without trunk ports. In order for them to ping one another, we could need to configure trunk ports.

Part 2: Configure Subinterfaces

Step 1: Configure subinterfaces on R1 using the 802.1Q encapsulation.

- Create the subinterface G0/0.10.
 - Set the encapsulation type to 802.1Q and assign VLAN 10 to the subinterface.
 - Refer to the **Address Table** and assign the correct IP address to the subinterface.

```
R1(config)# int g0/0.10
```

```
R1(config-subif)# encapsulation dot1Q 10
```

```
R1(config-subif)# ip address 172.17.10.1 255.255.255.0
```

- Repeat for the G0/0.30 subinterface.

Step 2: Verify Configuration.

- Use the **show ip interface brief** command to verify subinterface configuration. Both subinterfaces are down. Subinterfaces are virtual interfaces that are associated with a physical interface. Therefore, in order to enable subinterfaces, you must enable the physical interface that they are associated with.
- Enable the G0/0 interface. Verify that the subinterfaces are now active.

Part 3: Test Connectivity with Inter-VLAN Routing

Step 1: Ping between PC1 and PC3.

From **PC1**, ping **PC3**. The pings should still fail. Explain.

The subinterfaces have been configured, but trunk ports are still required for PC1 and PC3 to communicate since they are on different networks.

Step 2: Enable trunking.

- On **S1**, issue the **show vlan** command.

What VLAN is G0/1 assigned to?

It is assigned to the default VLAN (1).

- Because the router was configured with multiple subinterfaces assigned to different VLANs, the switch port connecting to the router must be configured as a trunk. Enable trunking on interface G0/1.

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How can you determine that the interface is a trunk port using the **show vlan** command?

Trunk ports are not shown in the table displayed from the 'show vlan' command. G0/0 does not appear in that table.

- c. Issue the **show interface trunk** command to verify that the interface is configured as a trunk.
G0/0 is displayed as having been configured a trunk.

Step 3: Test Connectivity

If the configurations are correct, PC1 and PC3 should be able to ping their default gateways and each other.

What addresses do PC1 and PC3 use as their default gateway addresses?

```
show vlan
-----
VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/10, Fa0/12, Fa0/13, Fa0/14
                                           Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/2
10   VLAN0010                active    Fa0/11
30   VLAN0030                active    Fa0/6
1002 fddi-default            active
1003 token-ring-default      active
1004 fddinet-default         active
1005 trnet-default           active
```

```
VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
1    enet  100001    1500  -      -      -      -   -         0      0
10   enet  100010    1500  -      -      -      -   -         0      0
30   enet  100030    1500  -      -      -      -   -         0      0
1002 fddi  101002    1500  -      -      -      -   -         0      0
1003 tr   101003    1500  -      -      -      -   -         0      0
1004 fdnet 101004    1500  -      -      -      ieee -         0      0
1005 trnet 101005    1500  -      -      -      ibm  -         0      0
```

```
VLAN Type  SAID      MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
-----
```

Remote SPAN VLANs

```
-----
Primary Secondary Type      Ports
-----
```

```
S1#show in
S1#show interface tr
S1#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,30

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,30
```

PC1: 172.17.10.1

PC3: 172.17.30.1

The PCs use the R1 subinterface IPs as their default gateways.